

**Basic Principles of Medicine 2**  
**Module: Nervous System and Behavioral Sciences | Lecture 12**

# **Cranial Nerve IX & X**

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# Objectives

|                                |                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------|
| SOM.MKII.BPM2.1.NB.1.ANAT.0175 | List the names and give the roman numerical assigned to cranial nerves nine to twelve      |
| SOM.MKII.BPM2.1.NB.1.ANAT.0176 | Describe the boundaries of the posterior cranial fossa and the positioning of the foramina |
| SOM.MKII.BPM2.1.NB.1.ANAT.0177 | Give the exit foramen for the glossopharyngeal nerve and state which bone it is found in   |
| SOM.MKII.BPM2.1.NB.1.ANAT.0178 | Describe the types of nerve fibers found in the glossopharyngeal nerve                     |
| SOM.MKII.BPM2.1.NB.1.ANAT.0179 | Describe the branches and distribution of the glossopharyngeal nerve                       |
| SOM.MKII.BPM2.1.NB.1.ANAT.0180 | Describe the extracranial course of the glossopharyngeal nerve and what it supplies        |
| SOM.MKII.BPM2.1.NB.1.ANAT.0181 | Give the location of the nuclei of the glossopharyngeal nerve                              |
| SOM.MKII.BPM2.1.NB.1.ANAT.0182 | Give the name and location of the ganglia formed by the glossopharyngeal nerve             |
| SOM.MKII.BPM2.1.NB.1.ANAT.0183 | Explain the consequence of extracranial lesions of the glossopharyngeal nerve              |
| SOM.MKII.BPM2.1.NB.1.ANAT.0184 | Give the exit foramen for the vagus nerve and state which bone it is found in              |
| SOM.MKII.BPM2.1.NB.1.ANAT.0185 | Describe the types of nerve fibers found in the vagus nerve                                |
| SOM.MKII.BPM2.1.NB.1.ANAT.0186 | Describe the branches and distribution of the vagus nerve                                  |
| SOM.MKII.BPM2.1.NB.1.ANAT.0187 | Describe the extracranial course of the vagus nerve and what it supplies                   |
| SOM.MKII.BPM2.1.NB.1.ANAT.0188 | Give the location of the nuclei of the vagus nerve                                         |
| SOM.MKII.BPM2.1.NB.1.ANAT.0189 | Give the name and location of the ganglia formed by the vagus nerve                        |
| SOM.MKII.BPM2.1.NB.1.ANAT.0190 | Explain the consequence of extracranial lesions of the vagus nerve                         |
| SOM.MKII.BPM2.1.NB.1.ANAT.0203 | Describe the afferent and efferent limbs of the following reflexes: cough and gag.         |



## Recommended readings

Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 4th ed. Elsevier. Chapter 8.

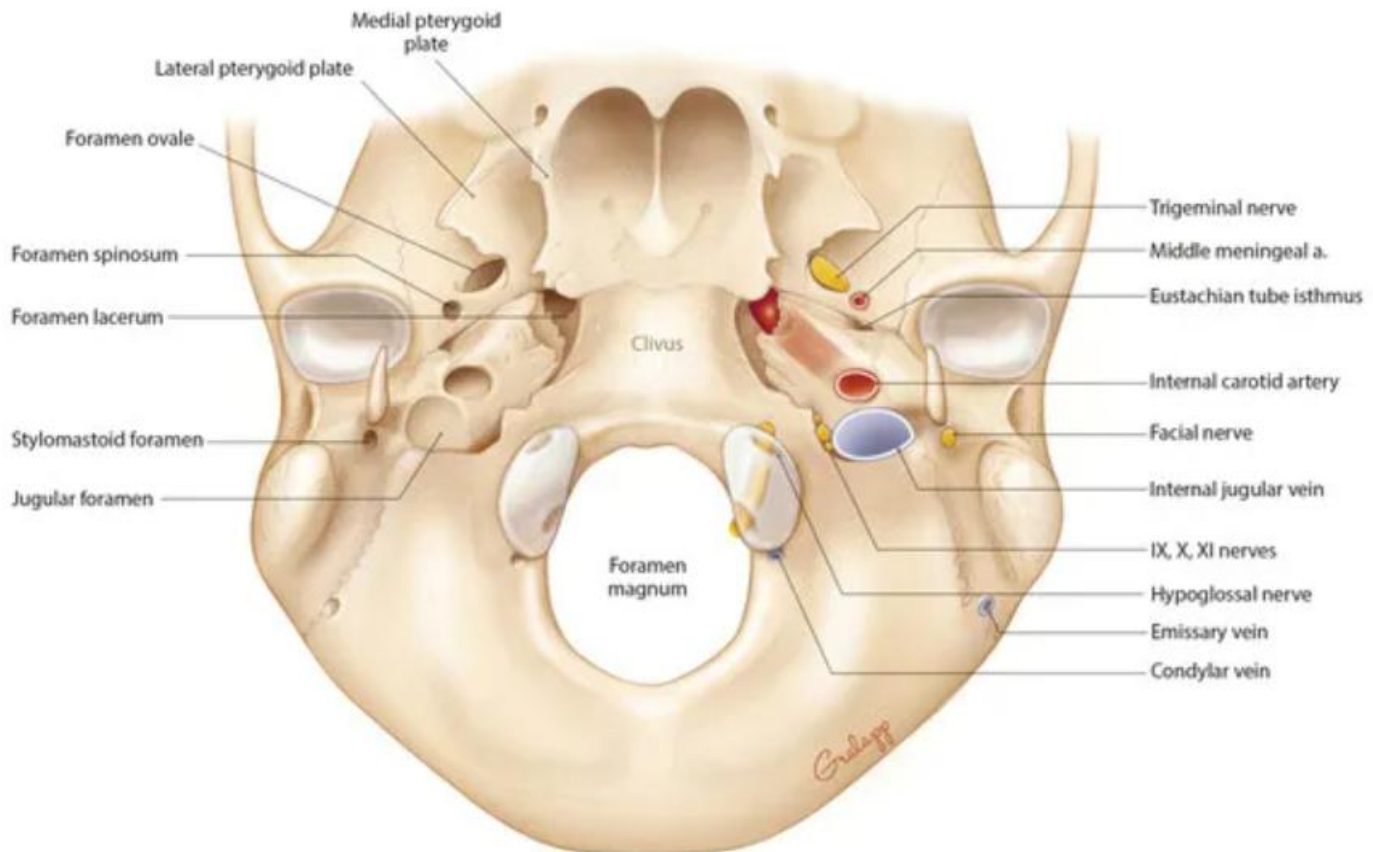
Siegel and Sapru Essential Neuroscience 4th ed. Chapter 13.

Neuroanatomy in clinical context. An atlas of structures, sections, systems and syndromes. Haines DE. 10th ed. Wolters Kluwer. Chapter 3

Pay special attention to the green 'In the Clinic' boxes which contain the clinical correlates for this topic.

# Overview

- *Cranial nerves: IX, X*
- *Glossopharyngeal nerve and Vagus nerve*
  - *Nuclei*
  - *Ganglia*
  - *Branches*



## To understand the cranial nerves discussed in this lecture, know:

- Location, name and type of **nucleus**
- **Attachment** to the brainstem/brain
- Where does it **exit (or enter)** the skull
- **Intracranial/extracranial pathway** (if applicable)
- Structures that it **supplies** along its path and its functional modality
- *What would be the **deficiency** if damaged at each of the above locations*



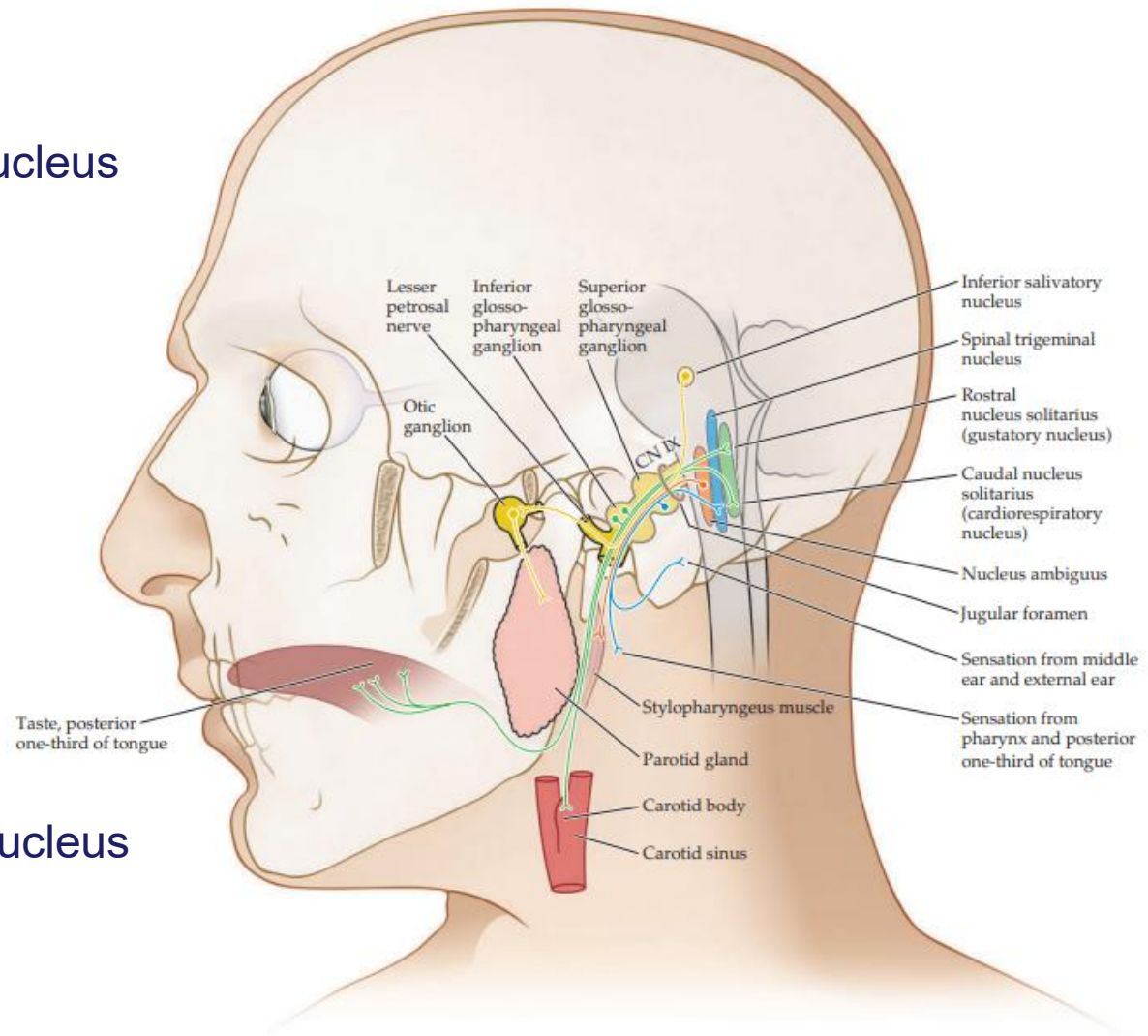
# Glossopharyngeal nerve - Overview

## Glossopharyngeal Nerve

|                         |                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | IX                                                                                                                                                        |
| Nuclei                  | <u>Sensory</u> : nucleus solitarius, spinal trigeminal nucleus<br><u>Motor</u> : nucleus ambiguus<br><u>Parasympathetic</u> : inferior salivatory nucleus |
| Attachment to brainstem | Medulla oblongata posterior to the olive                                                                                                                  |
| Major branches          | Stylopharyngeus (motor)<br>Carotid body and sinus<br>Tympanic plexus (sensory)<br>Posterior tongue (sensory)<br>Pharyngeal plexus (sensory)               |
| Ganglia                 | Superior (jugular) and Inferior (petrosal)                                                                                                                |

# Glossopharyngeal nerve - Nuclei

- General sensory
  - Spinal trigeminal nucleus
- Special sensory
  - Nucleus solitarius
- Motor
  - Nucleus ambiguus
- Parasympathetic
  - Inferior salivatory nucleus

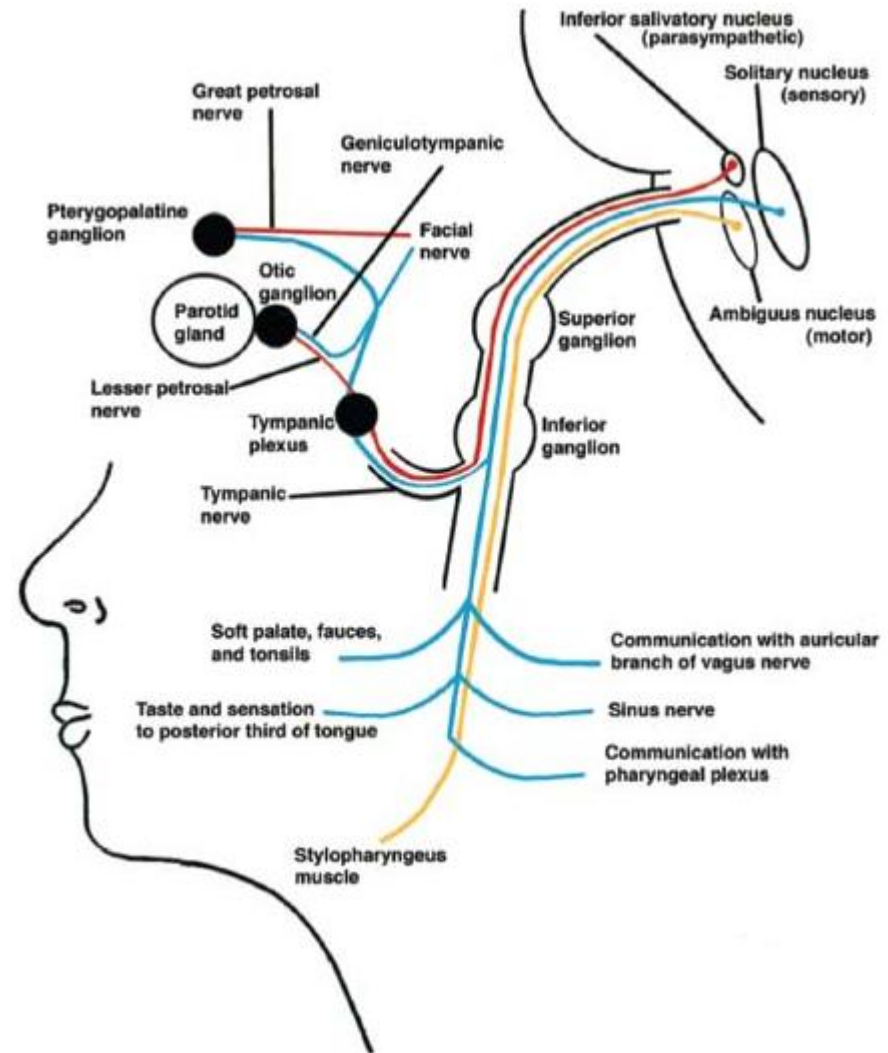


**FIGURE 12.20 Glossopharyngeal Nerve (CN IX)**  
Summary of glossopharyngeal nerve sensory and motor pathways.

# Glossopharyngeal nerve - Ganglia

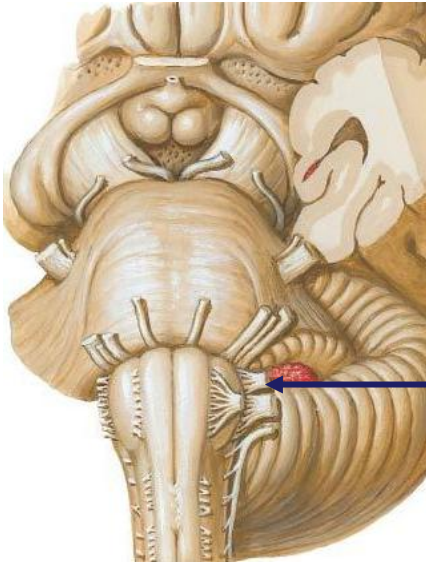
Two sensory ganglia:

- **Superior (jugular)**
  - General somatic sensation
- **Inferior (petrosal)**
  - General somatic sensation
  - General visceral sensation
  - Special visceral sensation

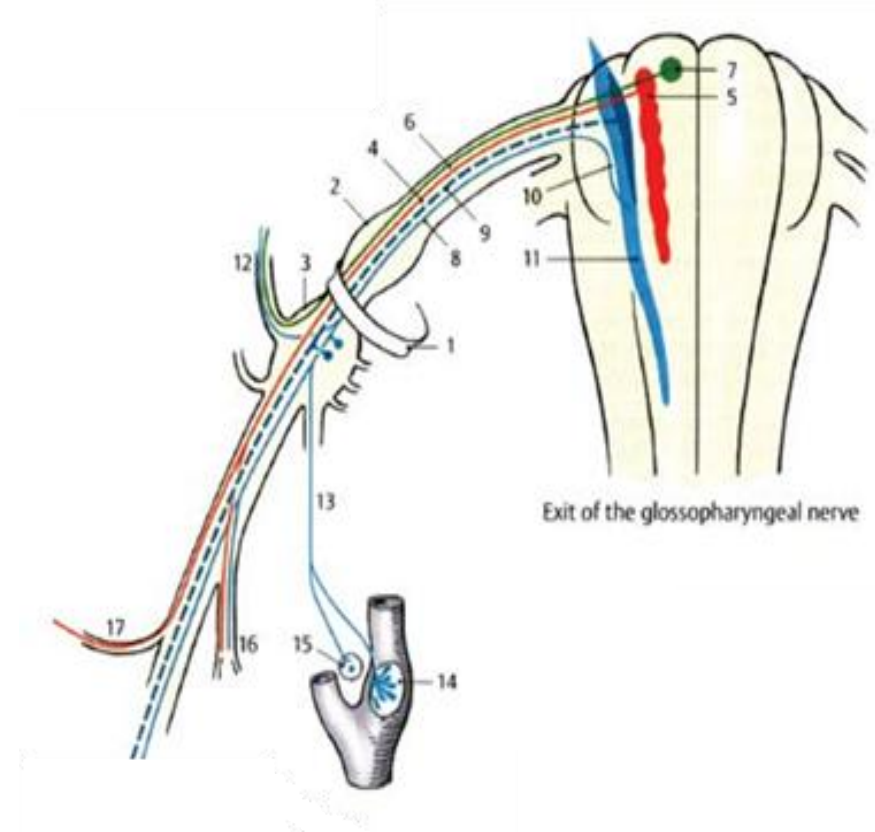


# Glossopharyngeal nerve – Initial Course

- Exits brainstem along upper ventrolateral medulla
  - between inferior olive and inferior cerebellar peduncle
- Exits cranial cavity via **jugular foramen**

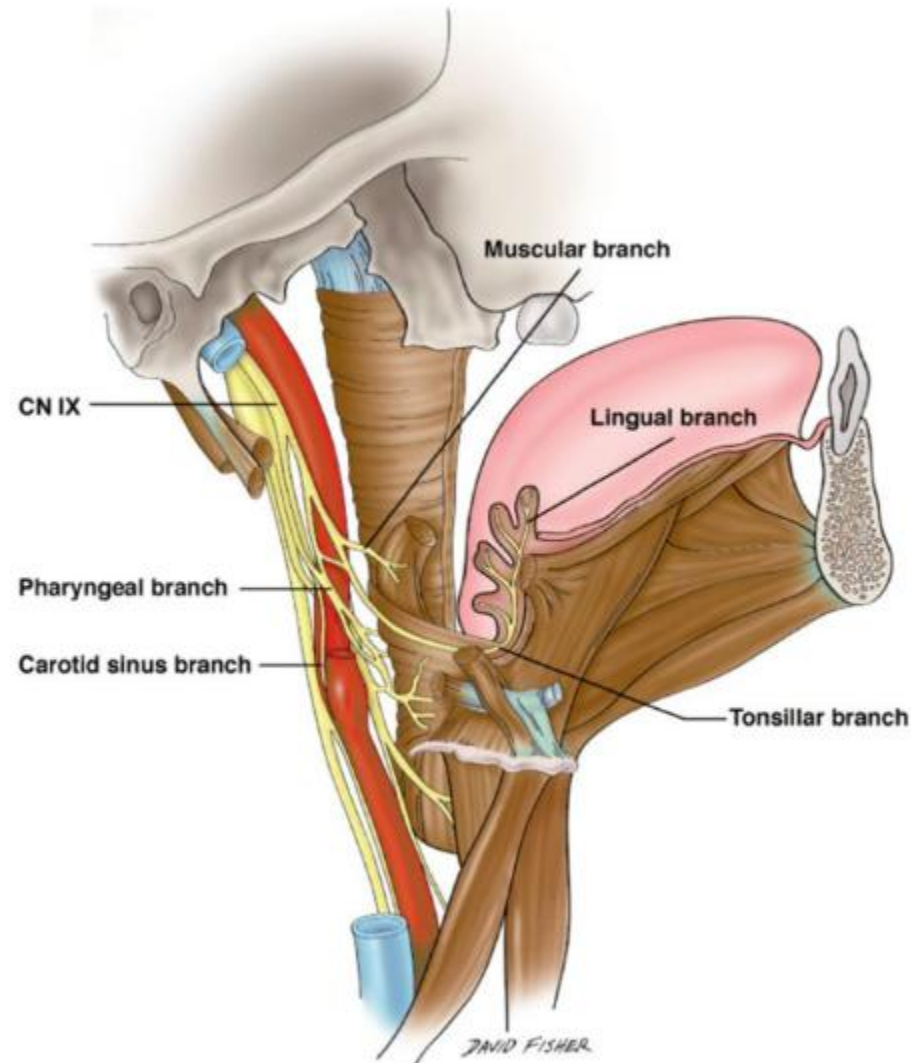


|          | Lateral       | Medial    | Lateral       |
|----------|---------------|-----------|---------------|
| Midbrain |               | III<br>IV |               |
| Pons     | V<br>VIII VII | VI        | V<br>VII VIII |
| Medulla  | IX<br>X       | XII       | IX<br>X       |



# Glossopharyngeal nerve – Extracranial Course

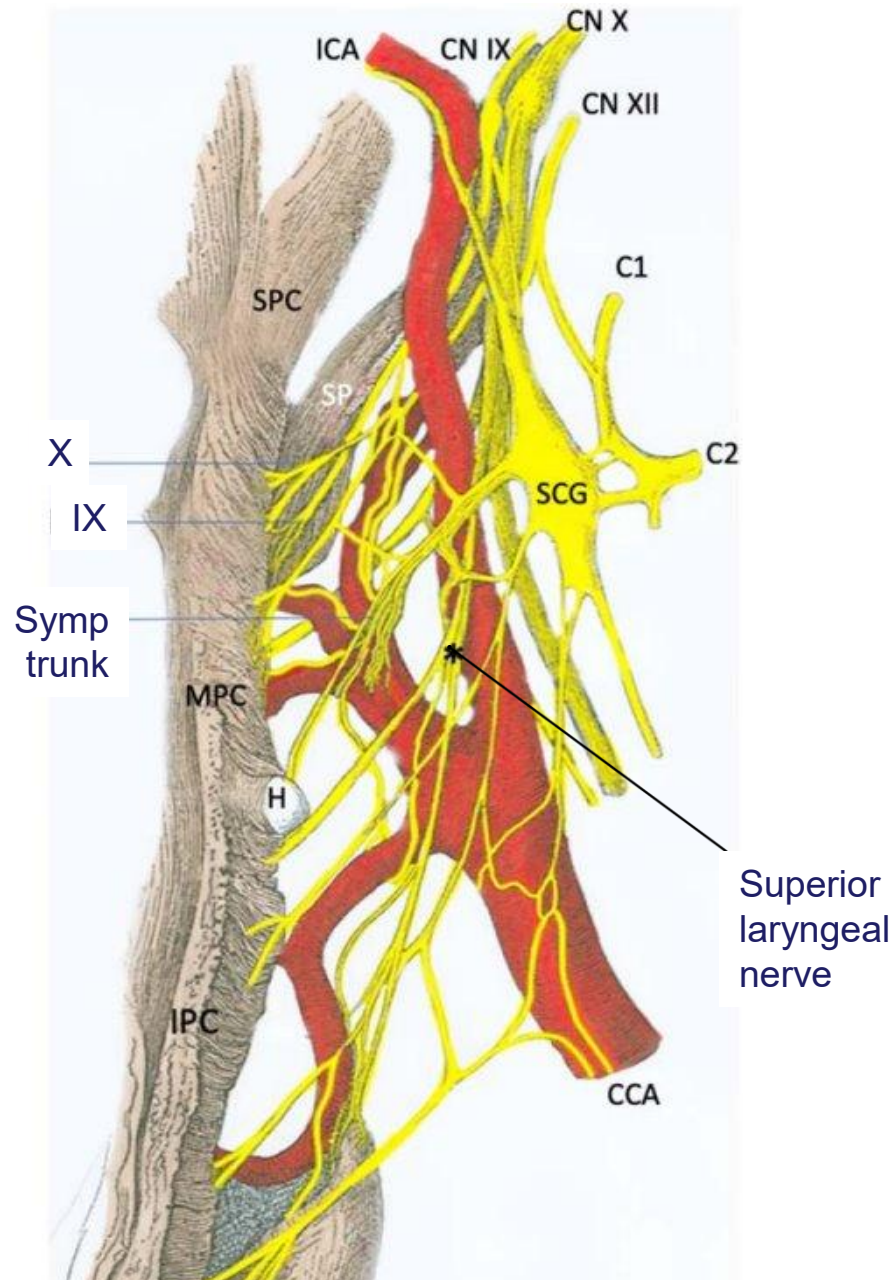
- Passes posterior to the carotid sheath
  - in close association with the cervical sympathetic chain
- Travels along the **stylopharyngeus muscle** in the tonsillar fossa
  - on its way to the posterior tongue
- Has branches that **join vagus branches**
  - form the **pharyngeal plexus**



# Glossopharyngeal nerve - Branches

## Carotid sinus nerve (of Hering):

- **Course**
  - Travels inferiorly while anterior to ICA
  - Communicates with X and sympathetic trunk
  - Bifurcates at division of ICA and ECA to supply carotid body and carotid sinus
- **Carotid sinus**
  - segment of dilated ICA
  - has baroreceptors that monitor blood pressure
- **Carotid body**
  - cluster of chemoreceptors in the adventitia of the carotid bifurcation
  - monitors O<sub>2</sub> pressure



# Glossopharyngeal nerve - Branches

- Sensory:

## Pharyngeal branch

- Combines with fibers of the vagus nerve to form the **pharyngeal plexus**
- innervates mucosa of oropharynx

## Tonsillar branch

- forms a network of nerves, known as the **tonsillar plexus**
- innervates the palatine tonsils

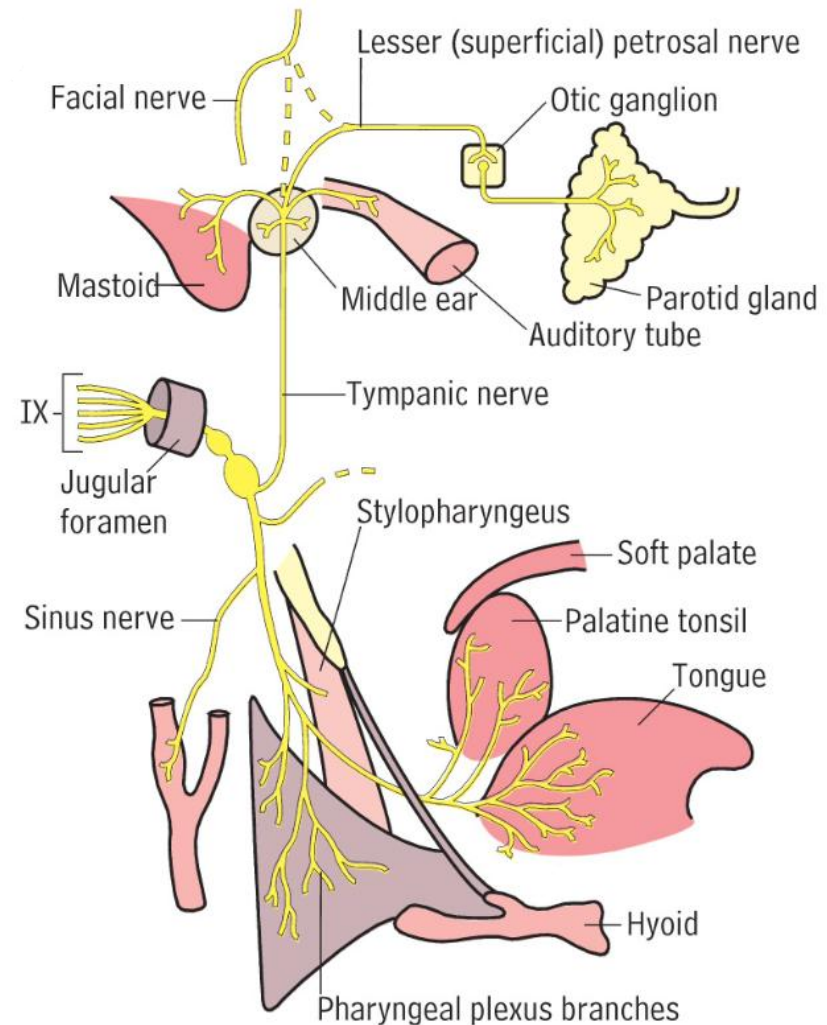
## Lingual branch

- provides general sensation and taste to the posterior 1/3 of the tongue

- Motor:

## Nerve to stylopharyngeus

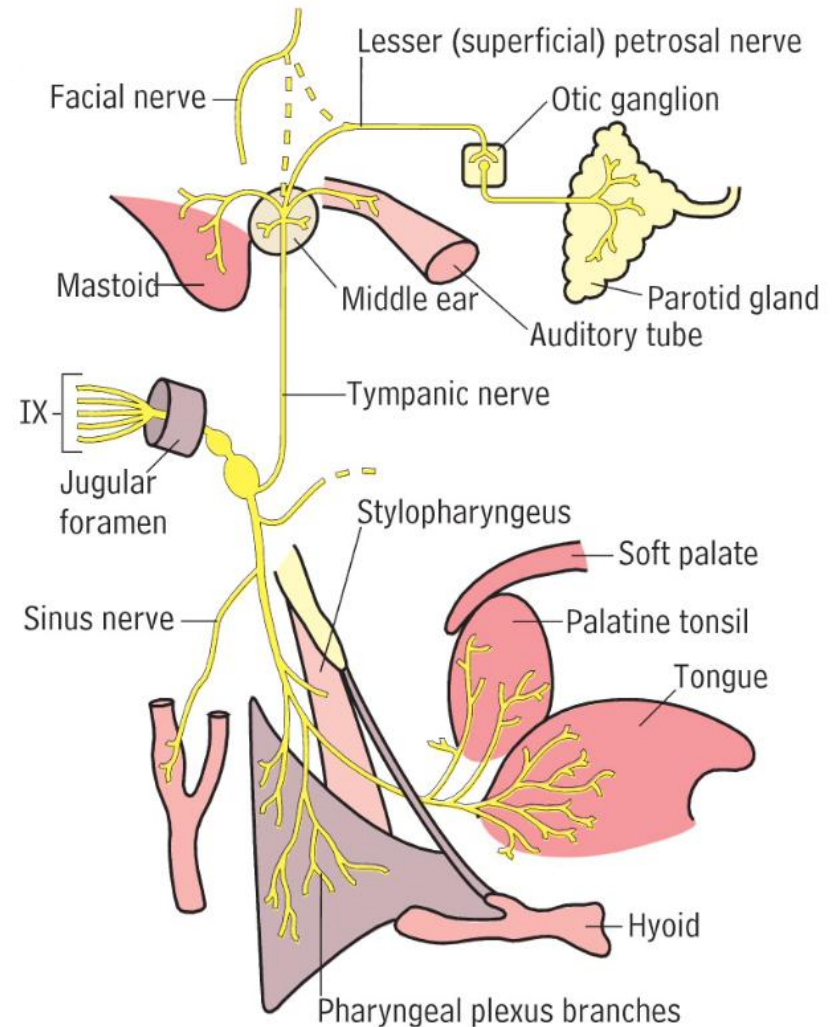
- elevates and shortens the pharynx, creating a dilated lumen for food to pass through



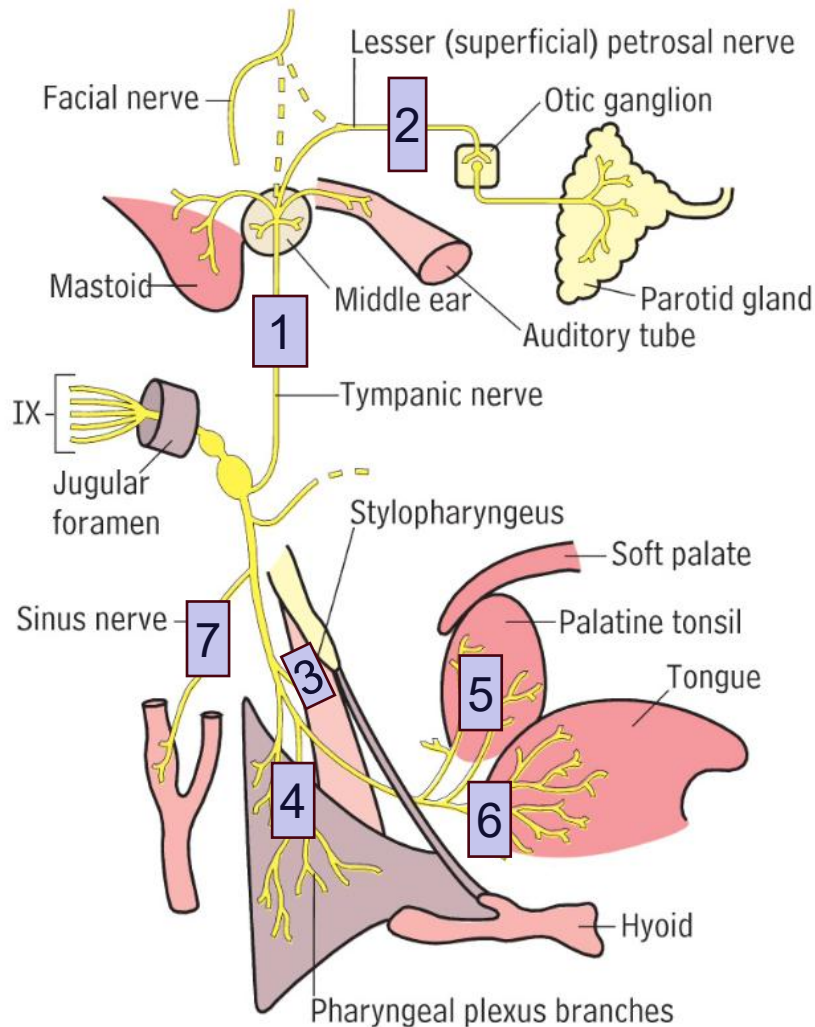
# Glossopharyngeal nerve - Branches

## Parasympathetic to Parotid:

- Fibers originate in inferior salivatory nucleus
- Travel with **tympanic nerve** (of Jacobson) to the middle ear
  - Supplies inner surface of tympanic membrane
- Form **tympanic plexus** in the tympanic cavity
- Enters middle cranial fossa as **lesser petrosal nerve**
- Exits cranium via foramen ovale and synapses in **otic ganglion**
- Postganglionic fibers are conveyed via **auriculotemporal nerve (V<sub>3</sub>)** to parotid
  - Auriculotemporal partially supplies external tympanic membrane



# Glossopharyngeal nerve – Lesions

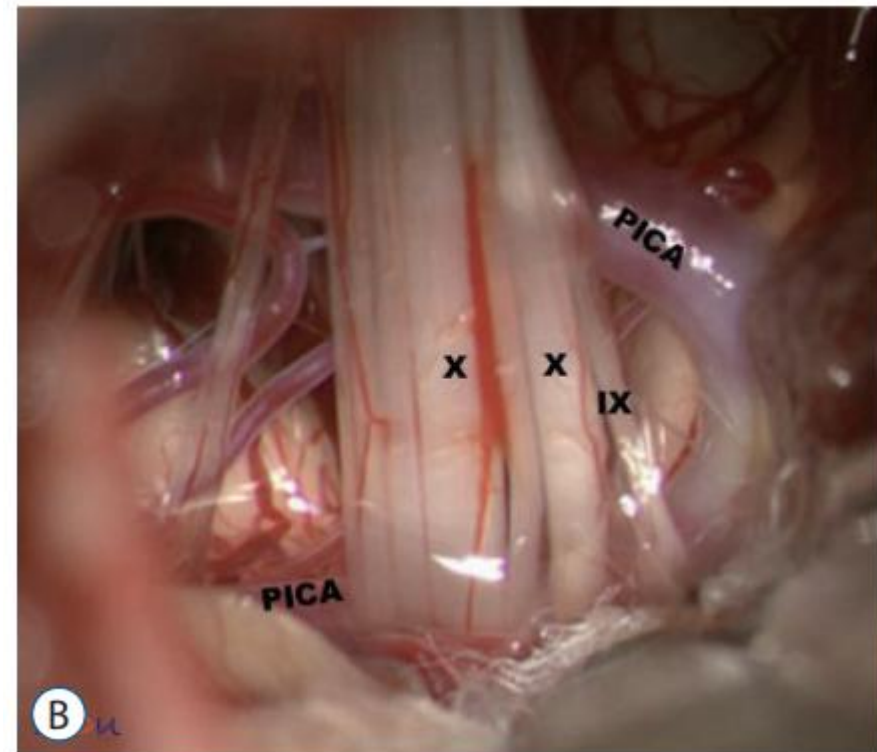


1. Loss of parasympathetic innervation to the parotid gland and sensation from the inner tympanic membrane
2. Loss of parasympathetic innervation to the parotid gland
3. Loss of motor fibers to stylopharyngeus muscle
4. Loss of sensation from mucosa of the oropharynx
5. Loss of sensation from mucosa of the palatine tonsil
6. Loss of special sensory (taste) and general sensation from the posterior one-third of the tongue
7. Loss of visceral sensory stimulus from the carotid bodies and sinus

# Clinical Correlate – Glossopharyngeal Neuralgia

- Caused by irritation of the glossopharyngeal nerve
- Can be triggered by chewing, coughing, laughing, speaking or swallowing.
- Presents with repeated episodes of severe pain in the tongue, throat, ear, and tonsils
- Lasts from a few seconds to a few minutes.
- Pain occurs in episodes and may be severe
- Associated pathology:
  - Neurovascular compression
  - Tumors at the base of the skull
  - Infections of the throat and mouth

from DOI: 10.3340/jkns.2022.0178



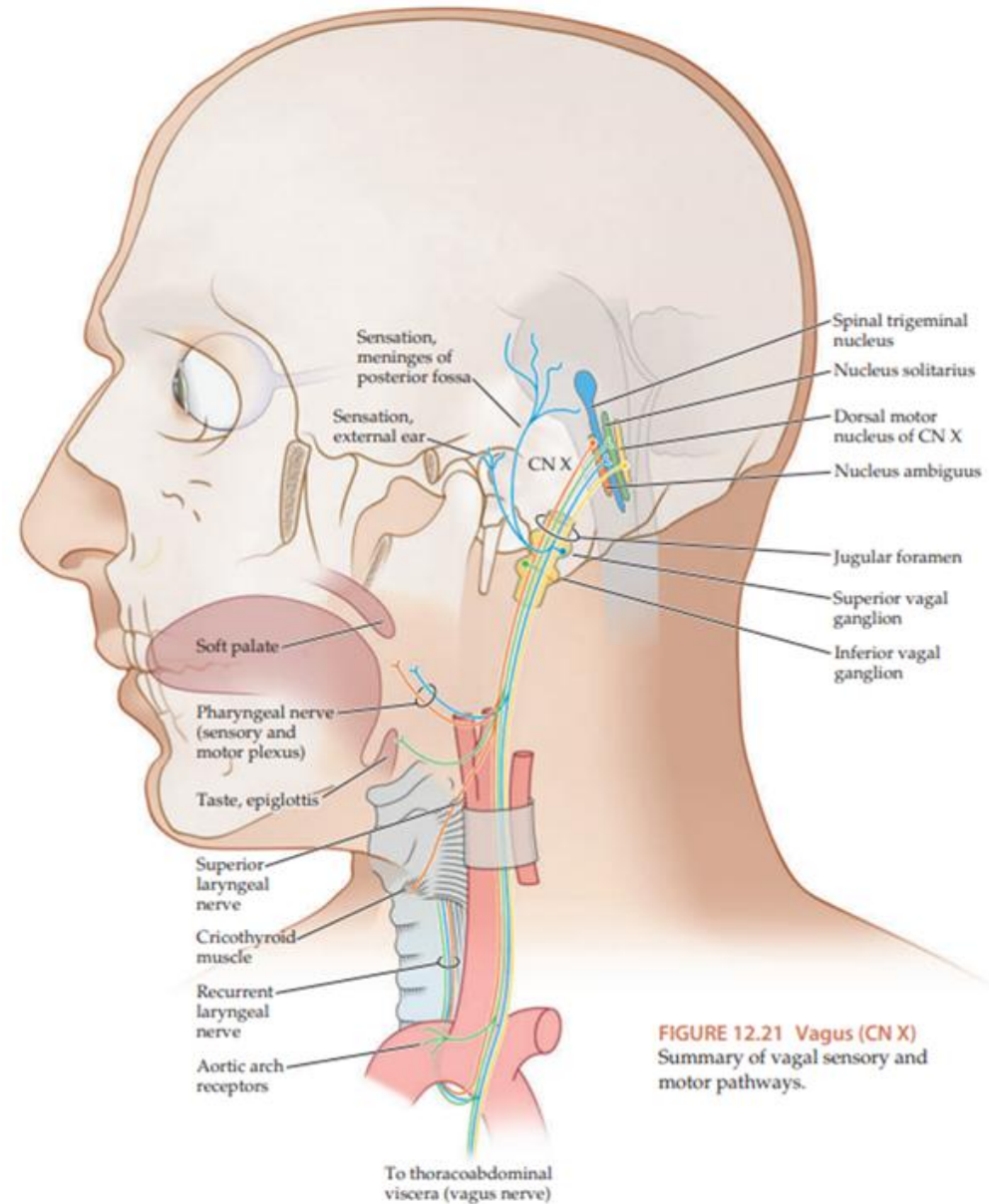
B : Tortuous PICA compressing the REZ and the cisternal portion of the CN IX-X complex.

# Vagus nerve - Overview

| Vagus Nerve             |                                                                                                                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | X                                                                                                                                                                                                                    |
| Nuclei                  | <u>Sensory</u> : nucleus solitarius, spinal nucleus<br><u>Motor</u> : nucleus ambiguus<br><u>Parasympathetic</u> : dorsal motor nucleus of vagus                                                                     |
| Attachment to brainstem | Medulla oblongata posterior to the olive                                                                                                                                                                             |
| Major branches          | Larynx, pharynx, soft palate (motor)<br>Larynx, pharynx, small/large intestine (parasymp)<br>Larynx, esophagus, bronchus, aortic body/sinus (sensory)<br>Pharyngeal plexus (sensory)<br>External ear canal (sensory) |
| Ganglia                 | Superior and Inferior                                                                                                                                                                                                |

# Vagus nerve - Nuclei

- General sensory
  - Spinal trigeminal nucleus
- Special sensory
  - (rostral) nucleus solitarius
- Motor
  - Nucleus ambiguus
- Parasympathetic
  - Dorsal motor nucleus

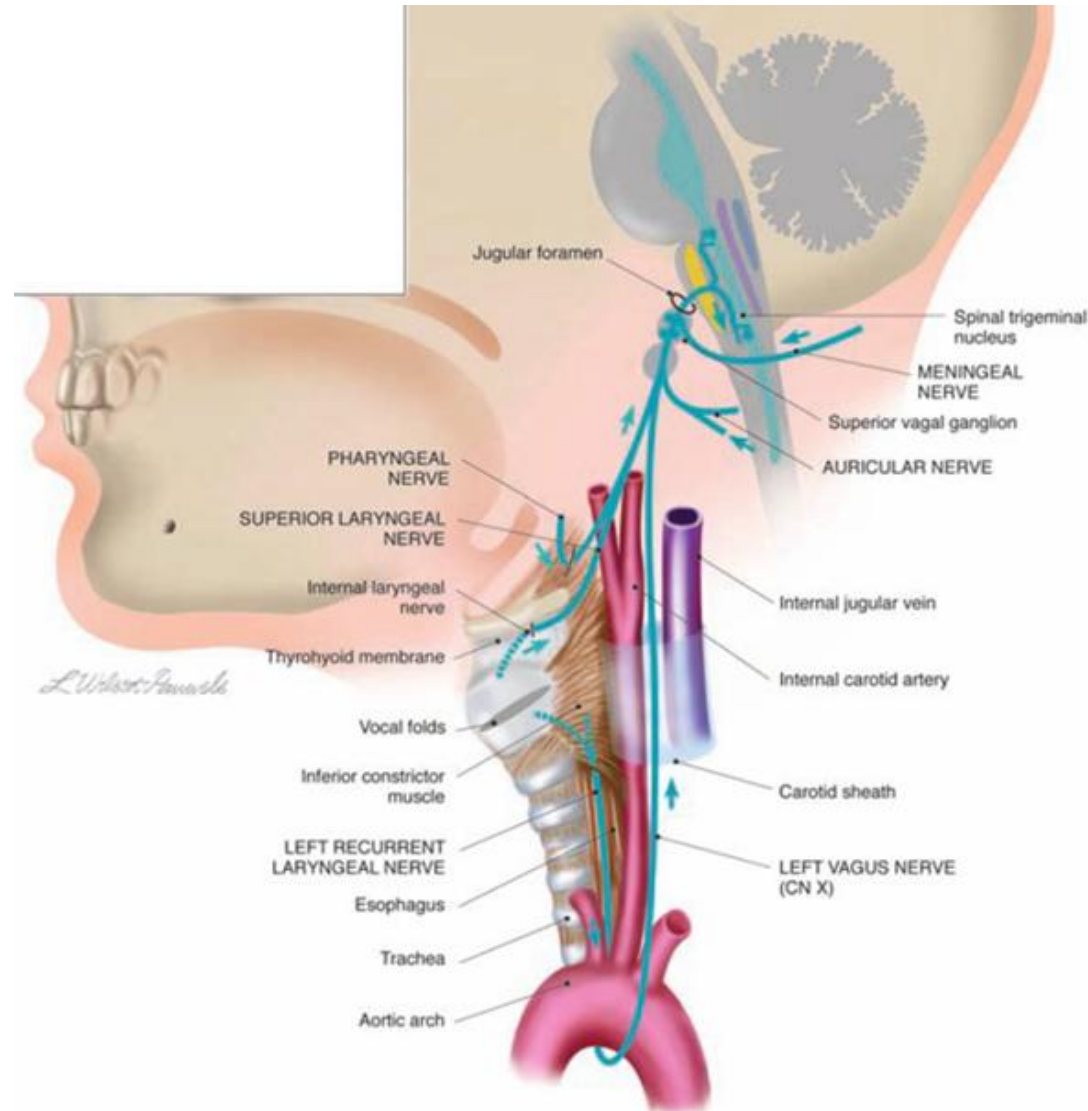


**FIGURE 12.21 Vagus (CN X)**  
Summary of vagal sensory and motor pathways.

# Vagus nerve - Ganglia

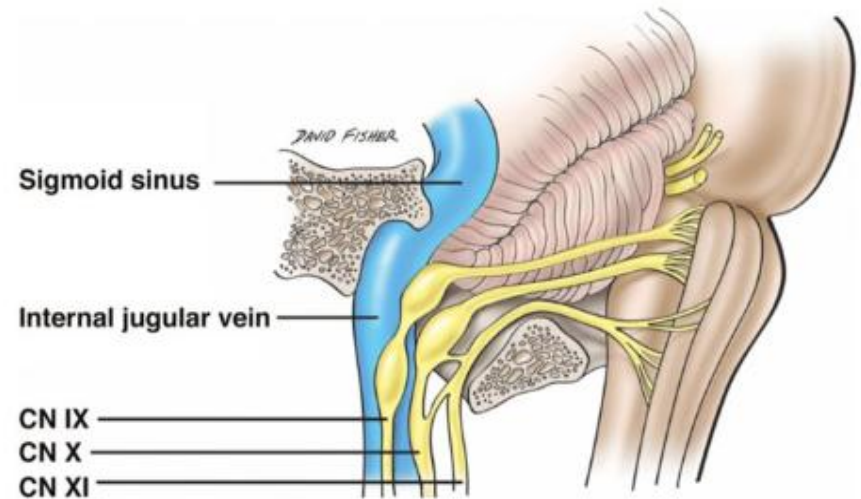
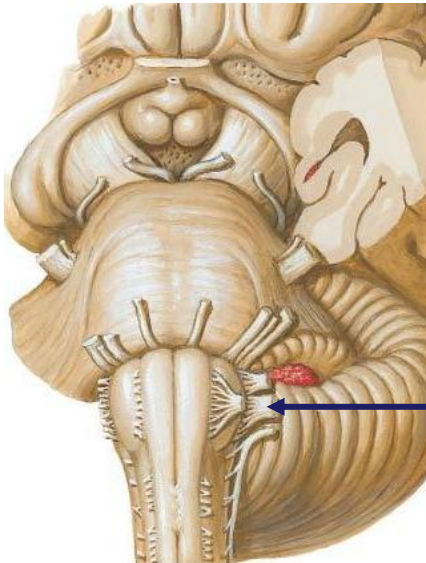
Two sensory ganglia:

- **Superior (jugular)**
  - General somatic sensation
- **Inferior (nodosal)**
  - General visceral sensation
  - Special visceral sensation



# Vagus nerve – Initial Course

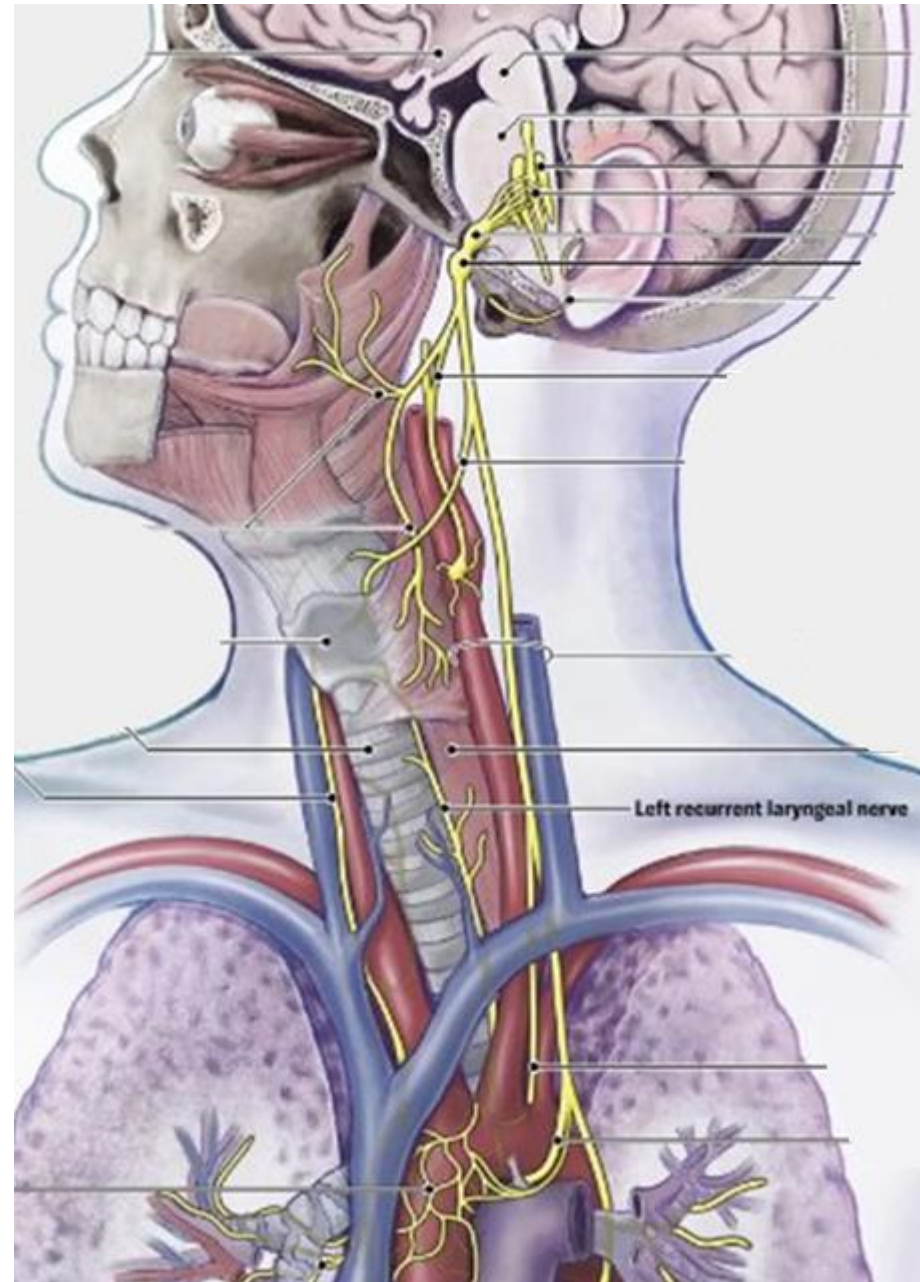
- Exits brainstem along upper ventrolateral medulla
  - Between inferior olive and inferior cerebellar peduncle
- Exits cranial cavity via **jugular foramen**



|          | Lateral       | Medial    | Lateral       |
|----------|---------------|-----------|---------------|
| Midbrain |               | III<br>IV |               |
| Pons     | V<br>VIII VII | VI        | V<br>VII VIII |
| Medulla  | IX<br>X       | XII       | IX<br>X       |

# Vagus nerve – Extracranial course

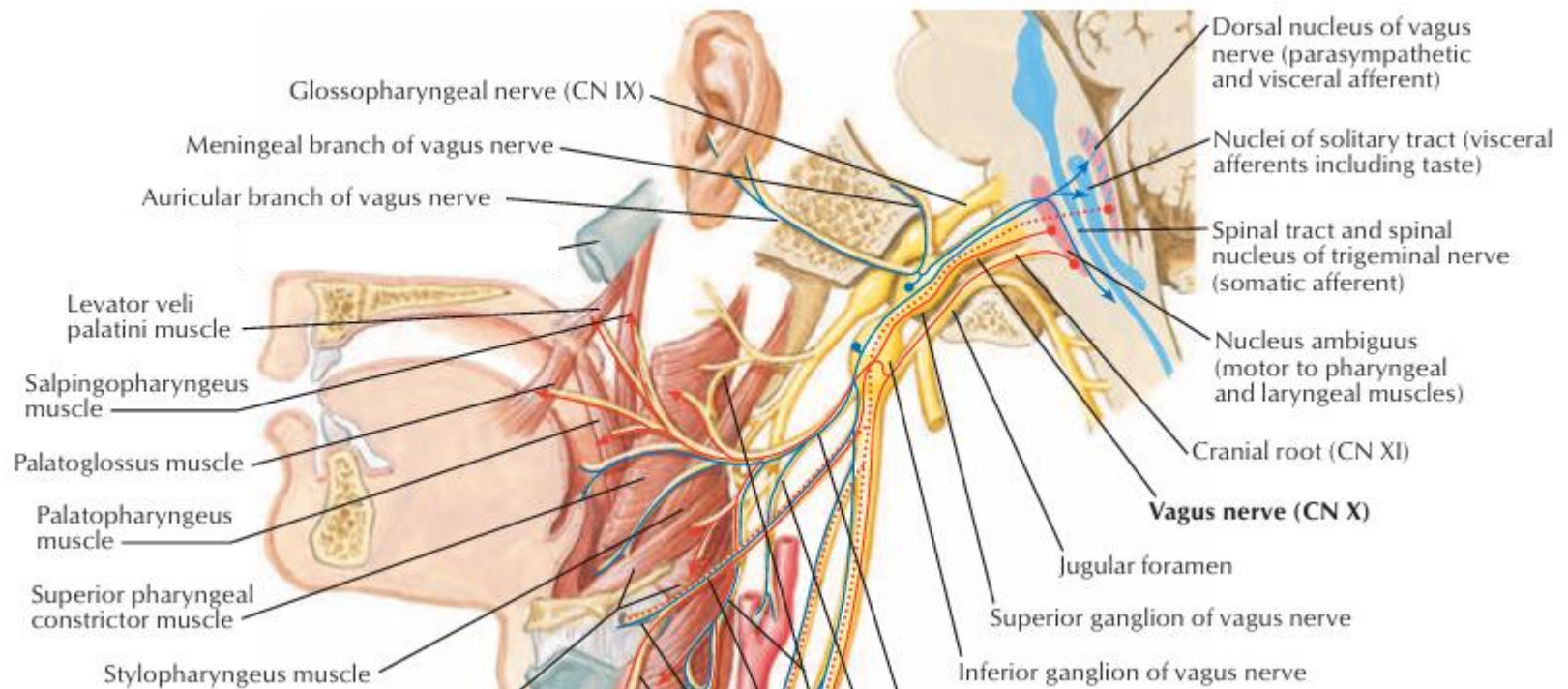
- Descends vertically through the neck
- Travels within the carotid sheath
- Lies posteriorly between common/internal carotid and internal jugular vein



# Vagus nerve - Branches

## Meningeal nerve

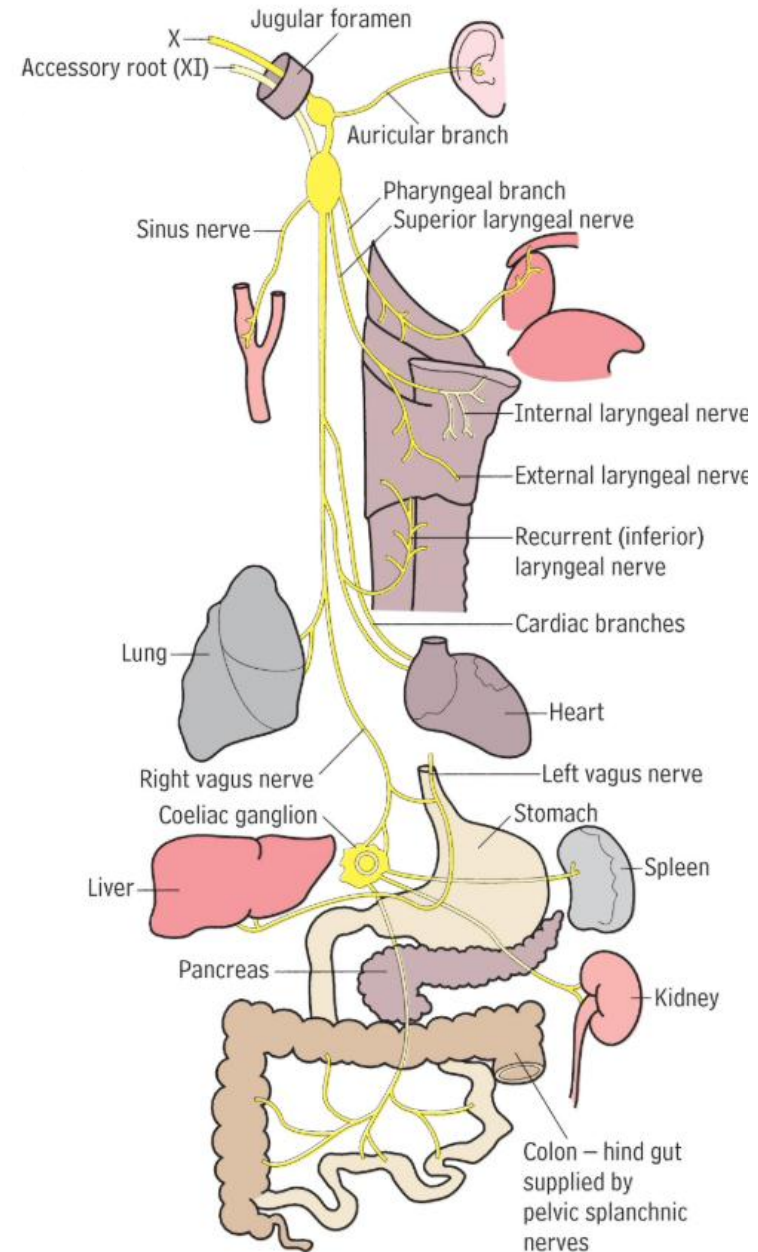
- arises at the at the superior ganglion
- re-enters the skull at the jugular foramen
- contains general afferent fibers
- supplies the dura of the posterior cranial fossa



# Vagus nerve - Branches

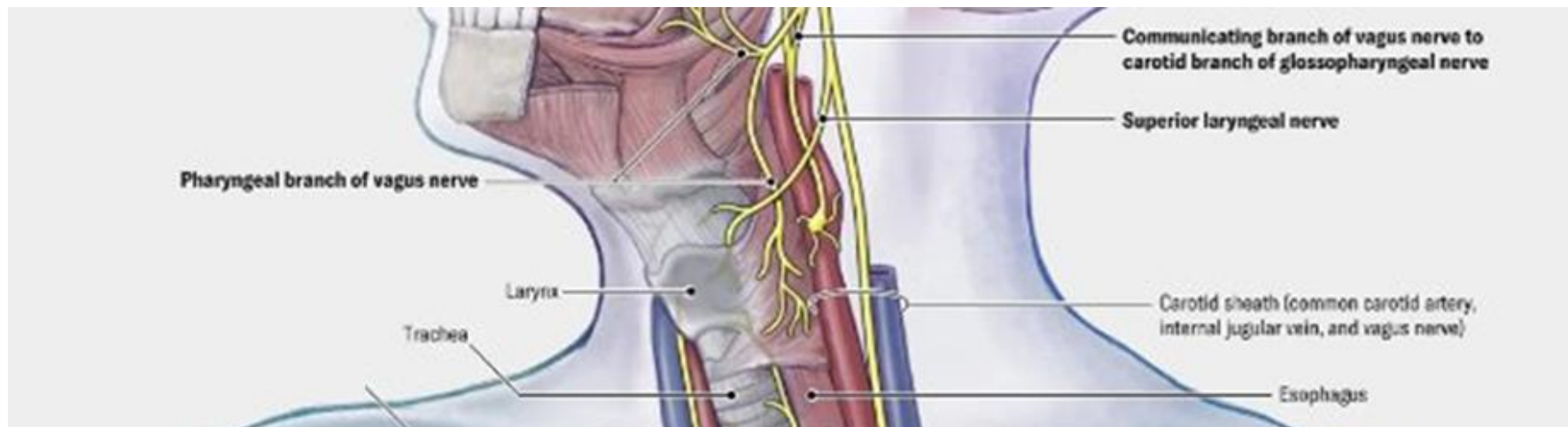
## Auricular/mastoid branch (Arnold's nerve)

- sensory innervation to:
  - external tympanic membrane
  - a portion of the posterior aspect of the external ear



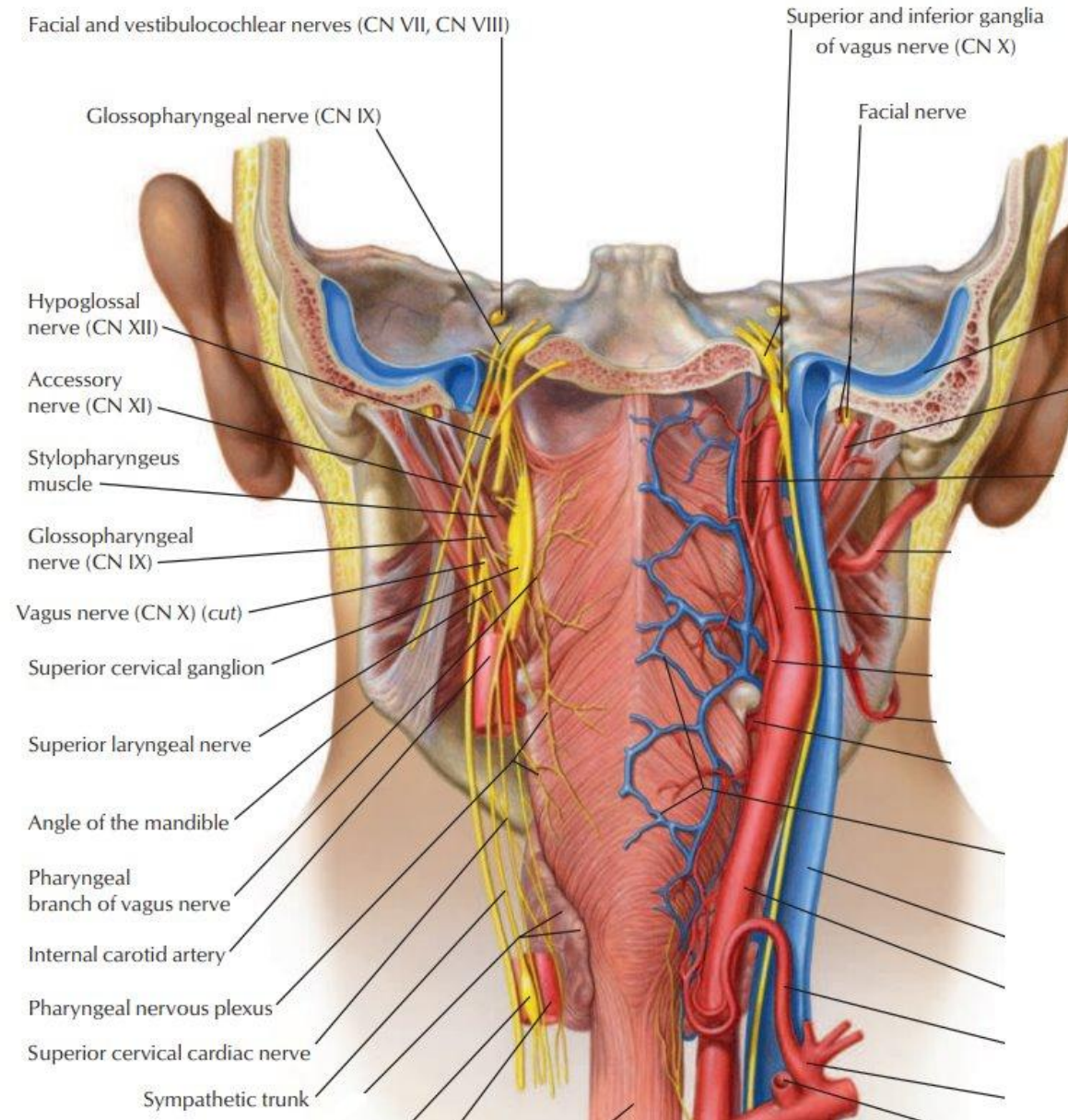
# Vagus nerve - Branches

- **Pharyngeal branch** supplies:
  - Pharyngeal plexus (motor)
    - Pharyngeal constrictor muscles: superior, middle, inferior
    - Longitudinal pharyngeal muscles
      - palatopharyngeus, salpingopharyngeus
  - Levator veli palatini
  - Palatoglossus
  - Uvula



# Pharyngeal Plexus

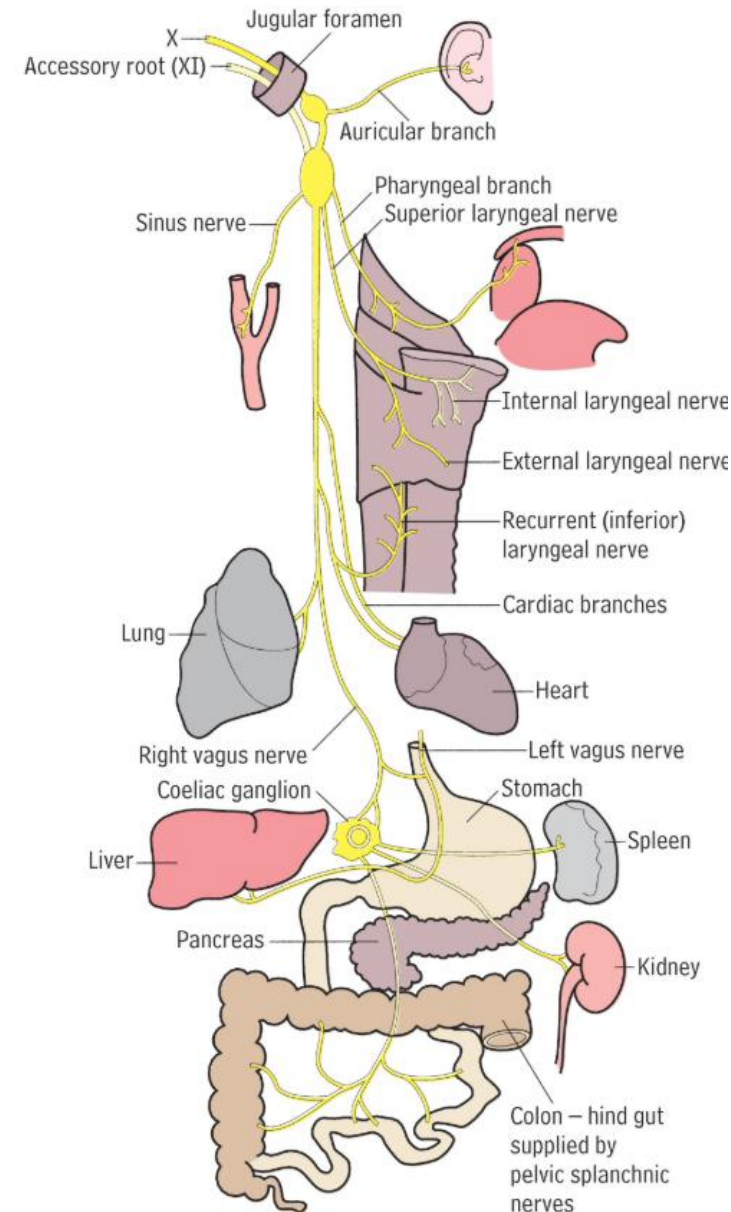
- **Superior cervical ganglion:** vasomotor
- **Sensory**
  - oropharynx: IX
  - laryngopharynx: X
- **Motor**
  - Stylopharyngeus: IX
  - All other muscles of the pharynx: X



# Vagus nerve - Branches

## Superior laryngeal nerve

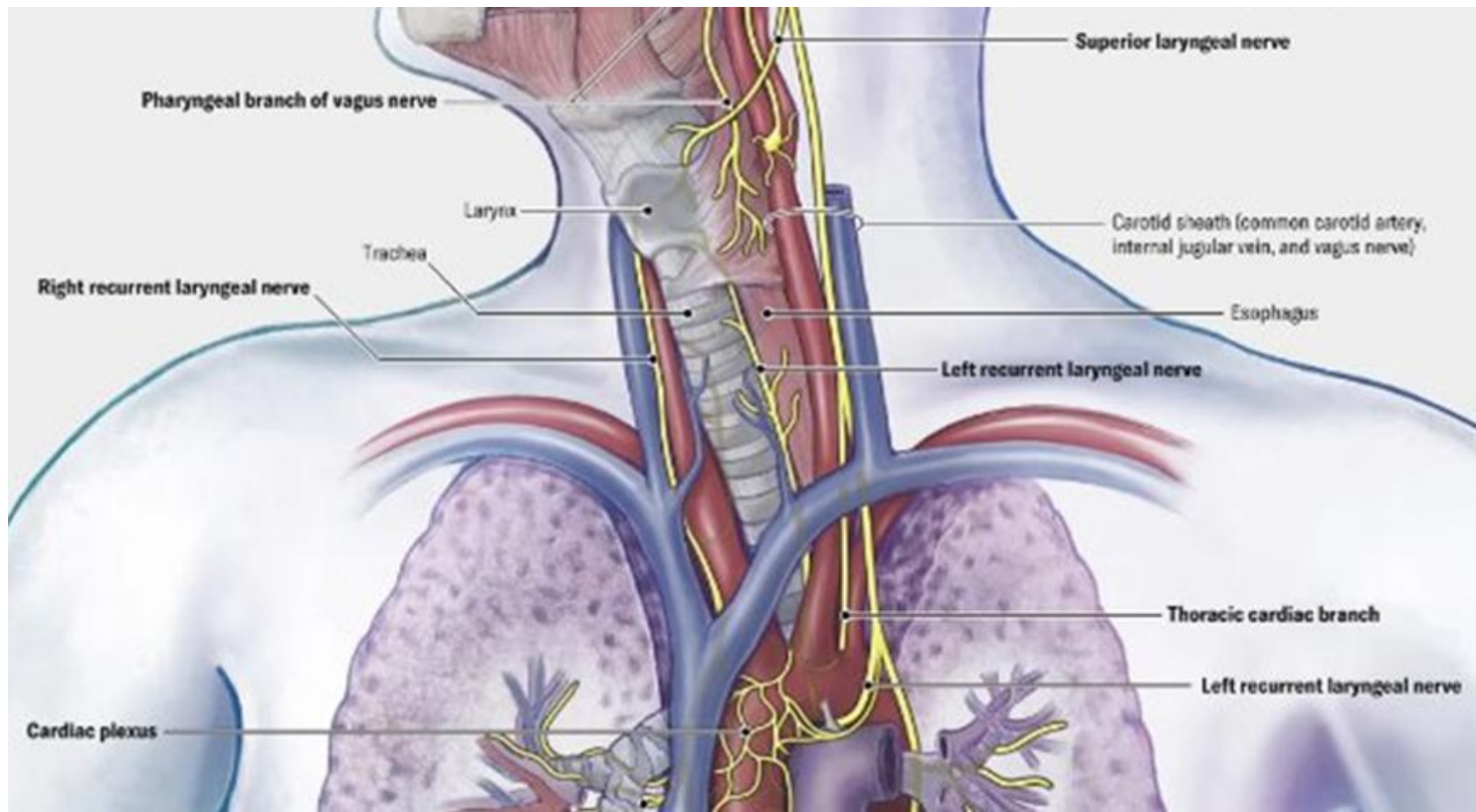
- Arises from the inferior ganglion
- Receives sympathetic fibers from the superior cervical ganglion
- Divides into external and internal branches at the tip of the hyoid bone.
- **External laryngeal branch**
  - Innervates cricothyroid
- **Internal laryngeal branch**
  - Enters larynx through thyrohyoid membrane
  - Sensory and secretomotor innervation to most of the mucosa above the glottis (vocal folds)



# Vagus nerve - Branches

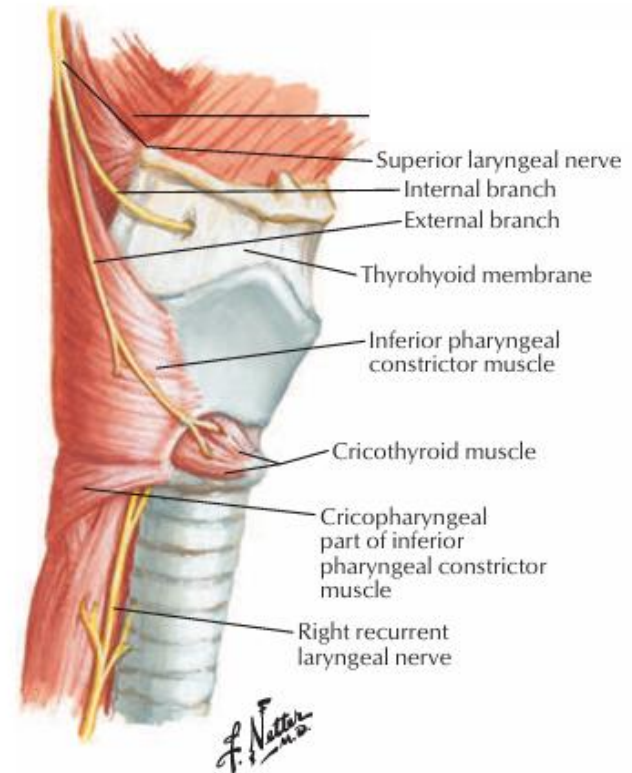
## Recurrent laryngeal nerve (inferior laryngeal nerves)

- Right : Loop around the right subclavian artery
- Left : Loops around the aortic arch
- Ascend along the tracheoesophageal groove and to the larynx.
- **Motor** innervation to all the intrinsic muscles of the larynx nerve, except *cricothyroid muscle*
- **Sensory and secretomotor** supply to the laryngeal mucosal structures below the level of the glottis (vocal cords ) and trachea

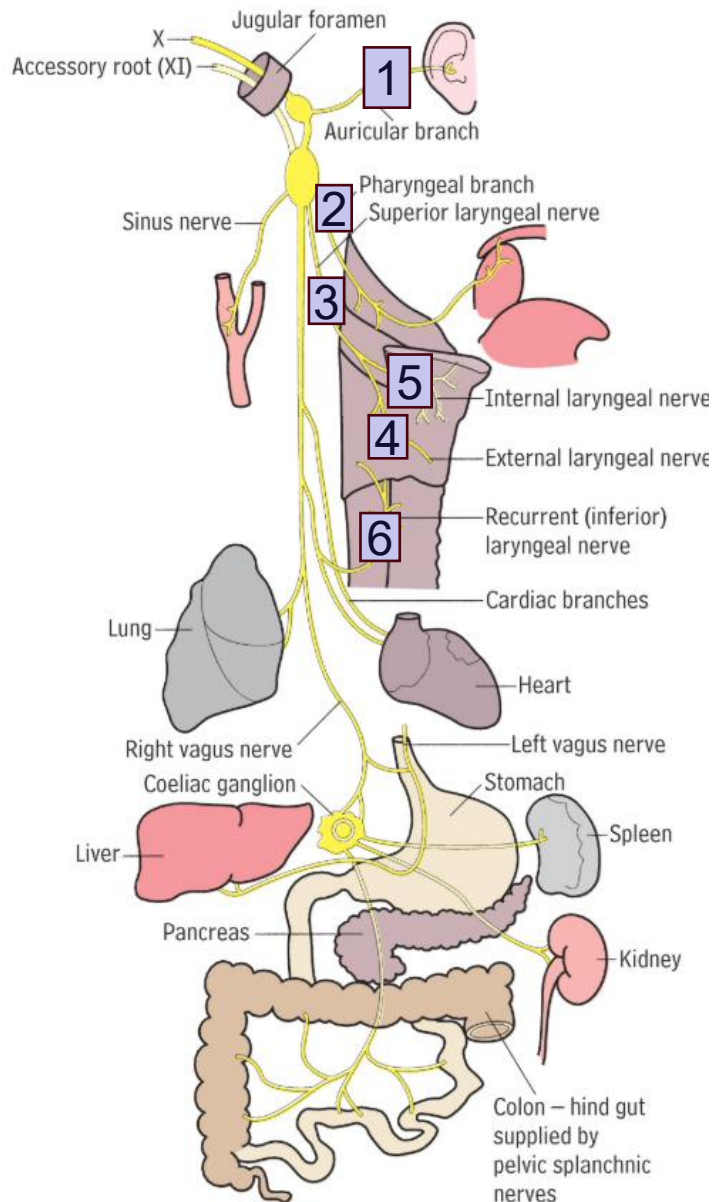


# Vagus nerve – Branches

| Muscle                            | Innervation                                    | Main Action(s)                                                      |
|-----------------------------------|------------------------------------------------|---------------------------------------------------------------------|
| Cricothyroid                      | External laryngeal nerve                       | Stretches and tenses vocal ligament                                 |
| Thyroarytenoid                    | Inferior laryngeal (recurrent laryngeal) nerve | Relaxes vocal ligament                                              |
| Posterior cricoarytenoid          |                                                | Abducts vocal folds                                                 |
| Lateral cricoarytenoid            |                                                | Adducts vocal folds                                                 |
| Transverse and oblique arytenoids |                                                | Adduct arytenoid cartilages                                         |
| Vocalis                           |                                                | Relaxes posterior vocal ligament while maintaining anterior tension |



# Vagus nerve - Lesions



1. Loss of sensory innervation from external tympanic membrane and portion of the posterior aspect of the external ear.
2. Loss of motor innervation to pharyngeal plexus
3. Loss of motor innervation to cricothyroid muscle and sensory and secretomotor innervation above glottis
4. Loss of motor innervation to cricothyroid muscle
5. Loss of sensory and secretomotor innervation above glottis
6. Loss of Motor innervation to all the intrinsic muscles of the larynx nerve, except *cricothyroid muscle* & loss of sensory and secretomotor supply to the laryngeal mucosal structures below the level of the glottis (vocal cords) and trachea

# Vagus nerve – Clinical Correlates

## Internal Superior laryngeal nerve palsy

### Common causes

- Surgical (thyroidectomy and parathyroidectomy)

### Symptoms:

- **Dysphagia** (difficulty swallowing)
- **Odynophagia** (painful swallowing)
- Sensation of a foreign body

## External Superior laryngeal nerve palsy

### Common causes

- Surgical (thyroidectomy and parathyroidectomy)

### Symptoms:

- **Reduced high-pitch voice**
- Hoarseness
- Voice fatigue
- Poor vocal volume and projection

## Recurrent nerve palsy

### Common causes

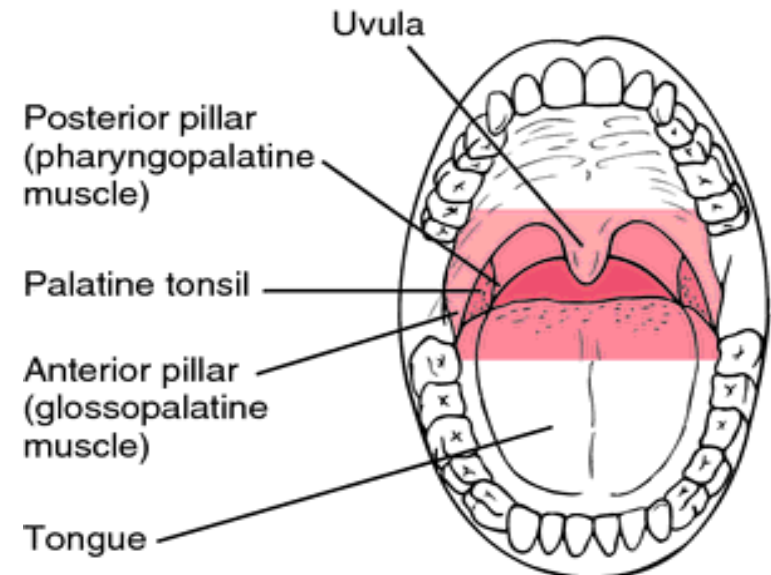
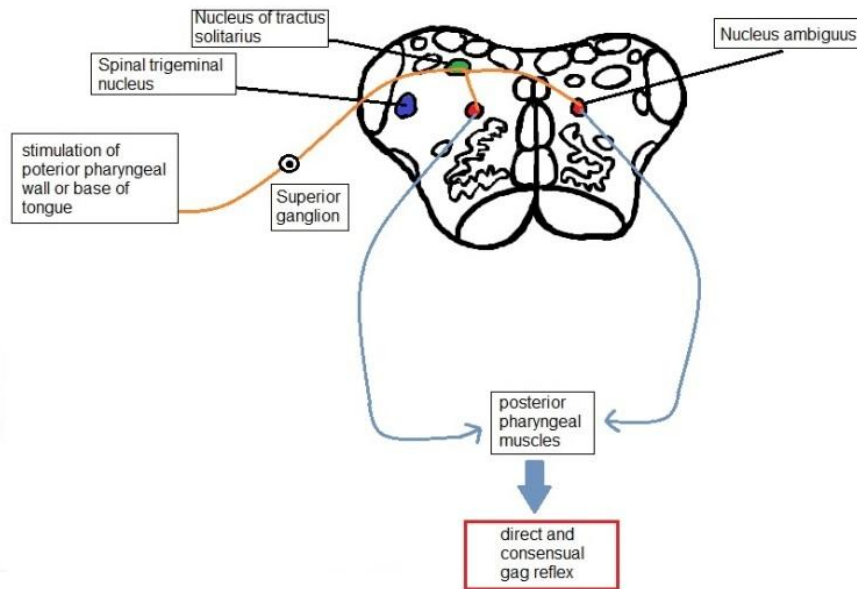
- Surgical (thyroidectomy and parathyroidectomy)
- Tumor
- Endotracheal intubation
- Viral illness

### Symptoms:

- **Unilateral:** New-onset hoarseness and noisy breathing
- **Bilateral:** Significant difficulties breathing and swallowing

# Clinical Correlate - Gag (Pharyngeal) Reflex

- Afferent limb of reflex (IX): evoked by stimulation of the posterior pharyngeal wall, tonsillar area, or base of the tongue
- Efferent limb of reflex (X): involuntary bilateral pharyngeal muscle contraction and elevation of the soft palate
  - Reflex can be classified as either somatogenic (physical touch) or psychogenic (thought of touch)
  - Absent reflex may be due to damage to the glossopharyngeal nerve



# Vagus nerve – Cough Reflex

- Protects lungs from aspiration and facilitates clearance of secretions and foreign bodies
- Afferent limb of reflex (X):
  - epithelium of the upper airways
  - diaphragm branches
- Efferent limb of reflex (X):
  - Nucleus ambiguus**: laryngeal muscles
  - Nucleus retroambiguus** (within ventral respiratory group): respiratory muscles

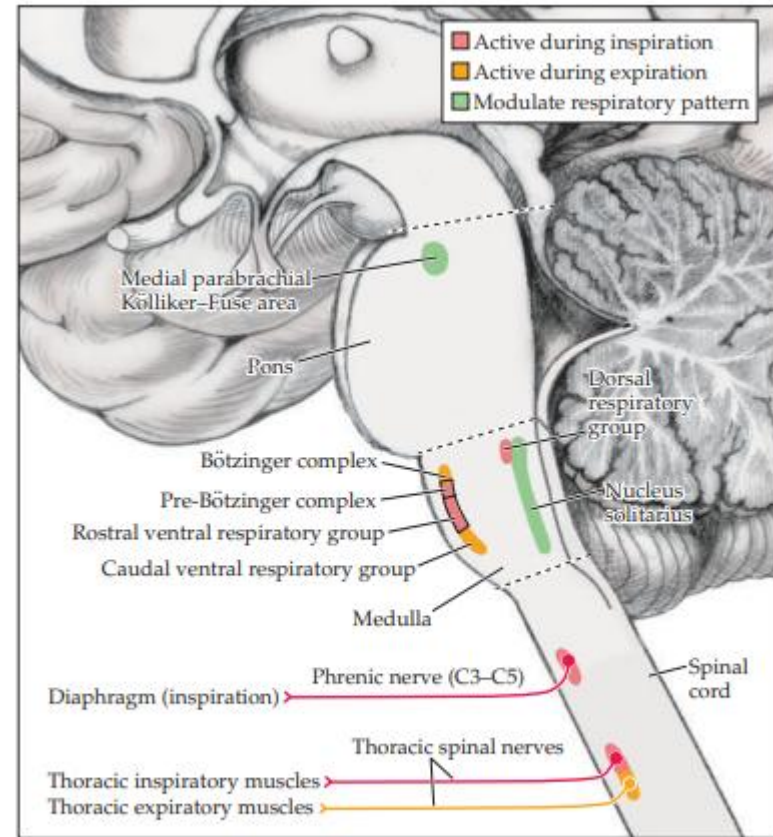
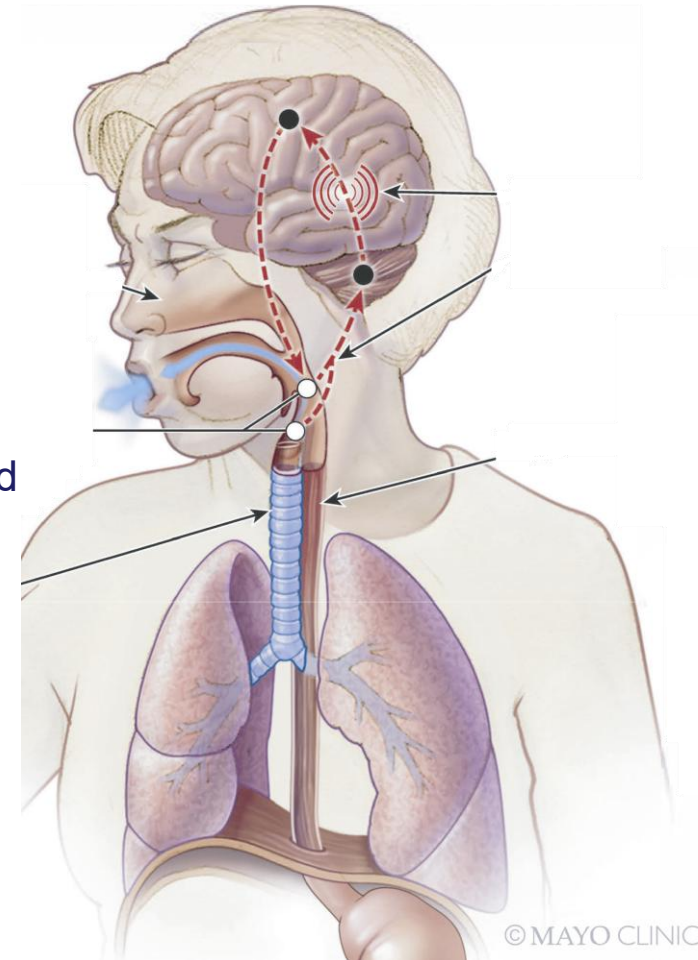


FIGURE 14.17 Brainstem Regions Involved in Respiratory Control

# Vagus nerve – Cough Reflex

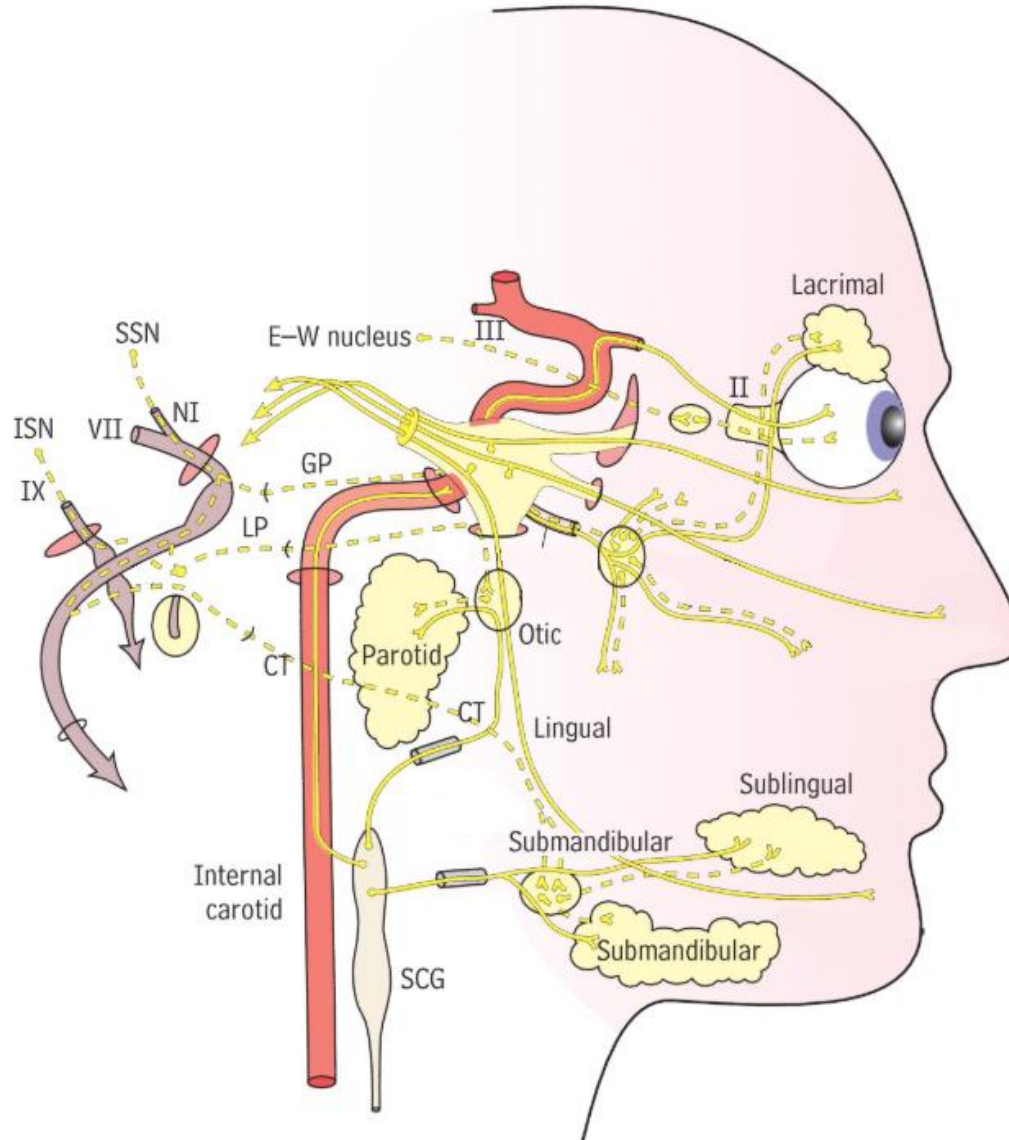
The mechanical events of a cough can be divided into three phases:

- 1) Inspiratory phase: Inhalation
  - generates the air volume necessary for an effective cough
- 2) Compression phase: Closure of the larynx combined with contraction of muscles of chest wall, diaphragm, and abdominal wall
  - results in a rapid rise in intrathoracic pressure
- 3) Expiratory phase: Opening of glottis opens and large airway compression
  - high expiratory airflow dislodges mucus from the airways and tracheobronchial tree



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# For Self-review: Head & Neck ANS Ganglia



- CT** chorda tympani
- E-W** Edinger–Westphal
- FO** foramen ovale
- FR** foramen rotundum
- GP** greater petrosal
- ISN** inferior salivatory nucleus
- LP** lesser petrosal
- NI** nervus intermedius
- SA** somatic afferent
- SCG** superior cervical ganglion
- SOF** superior orbital fissure
- SSN** superior salivatory nucleus

fin

